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General Knowledge
World Geography

STUDY NOTES

Geomorphology: Interior of the Earth

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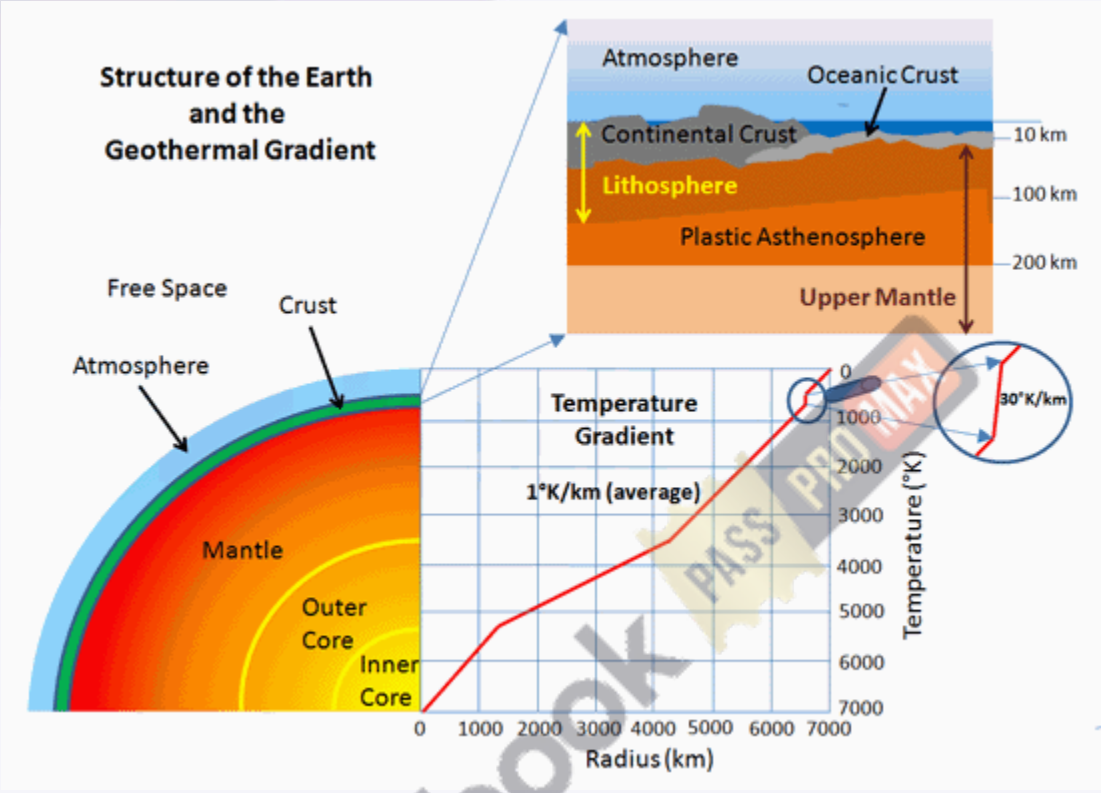
The structure of the earth is divided into three parts: a) Crust b) Mantle c) Core.



The Crust

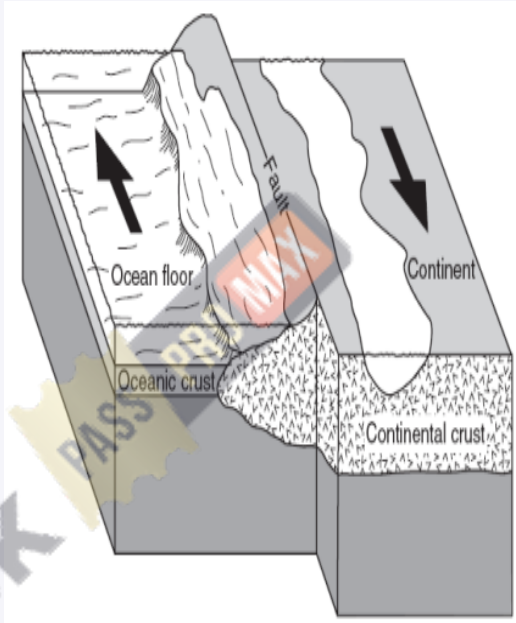
- The outermost solid part of the earth is called the **crust**.
- It is **brittle** in nature.
- The thickness of the crust varies under the oceanic and continental areas.

Oceanic crust	**Continental crust**
Composed mainly of **basaltic rocks**	Made of **granitic rocks**
Density ~3.0 g/cm³	**Density** ~2.7 g/cm³
Thinner- mean thickness of **5km**	**Thicker**-mean thickness **30km**
Younger/Newer	**Older**

- The **upper layer of the crust** is made up of **Silica and Aluminium (SiAl)** with a **density of 2.7g/cm3**.

- The **lower layer of the crust** is made up of **Silica and Magnesium(SiMa)** with a **density of 3g/cm3.**
- **SiAl and SiMa** together form Earth's Crust with an average **density of 3.0g/cm3.**
- The discontinuity which separates the upper crust from the lower crust is called **Conard Discontinuity.**
- The discontinuity that separates the Crust from the Mantle is called **mohorovicic discontinuity.**

Elements	Percentage
Oxygen	46.6%
Silicon	27.7%
Aluminium	8.1%
Iron	5.0%
Calcium	3.6%
Sodium	2.8%
Potassium	2.6%
Magnesium	2.1%
Others	1.5%



The Mantle

- **The portion beyond Moho discontinuity to a depth of 2900 km is called the Mantle.**
- The upper portion of the Mantle is called the **Asthenosphere**(up to 400km), Astheno means weak
- The crust and upper part of the mantle are called **Lithosphere**, ranging from 10-200km.
- The lower mantle extends below the Asthenosphere and is in a Solid state.
- **Repiti discontinuity** separates the upper mantle and the lower mantle.
- It is the source of Magma.
- **Density**: Increases with depth from about **3.3 to 5.7 g/cm³.**
- The mantle is formed of largely Silicate minerals rich in **Iron and Magnesium.**
- The transition zone between Mantle and core is called **Gutenberg Discontinuity.**

The core

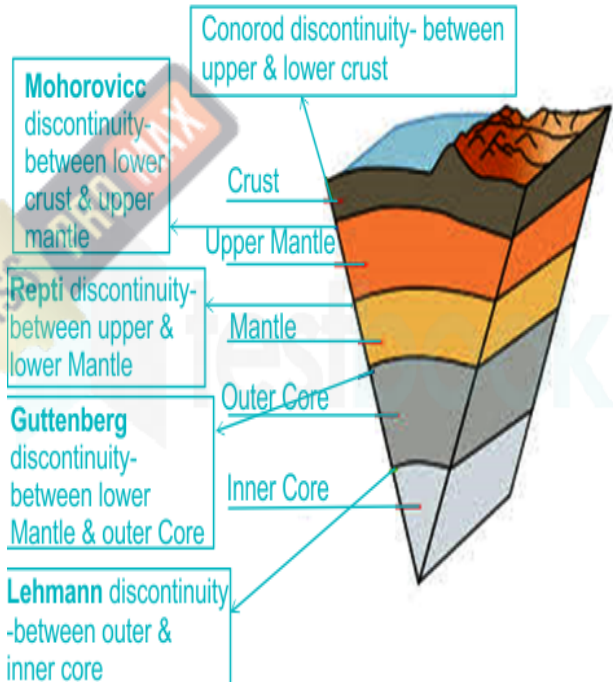
- The core is rich in **Nickel** and **Iron** that's why it is called the **NiFe layer.**
- **Lehman discontinuity separates the Upper Core from the Lower Core.**

Outer core (molten)	Inner core(Solid)
Thickness About 2,200 km.	**Thickness** About 1,220 km.
Mostly iron and nickel in a molten, **liquid** state.	**Solid** iron and nickel
Density: ~9.9 to 12.2 g/cm³.	**Density**: ~12.6 to 13.0 g/cm³.
Temperature: Estimated between 4,000°C to 6,000°C.	**Temperature**: Estimated to be around 5,400°C.

💡Points to remember💡

- **Conrad Discontinuity**: A boundary separating the upper crust from the lower crust.
- **Mohorovičić Discontinuity**: The boundary between the Earth's lower crust and the mantle.
- **Repetti Discontinuity:** A boundary between the upper mantle and lower mantle.
- **Gutenberg Discontinuity:** The boundary between the Earth's lower mantle and the outer core
- **Lehmann Discontinuity:** The boundary between the Earth's outer core and inner core.

The trick to remember: **Commander Mohit is ReGuLar (Co) (Mo) (Re)(Gu)(Le).**



- **Oxygen is most abundant Element in earth crust.**
- **Aluminium is most abundant metal in earth Crust.**
- **Iron is the most abundant metal in entire Earth.**